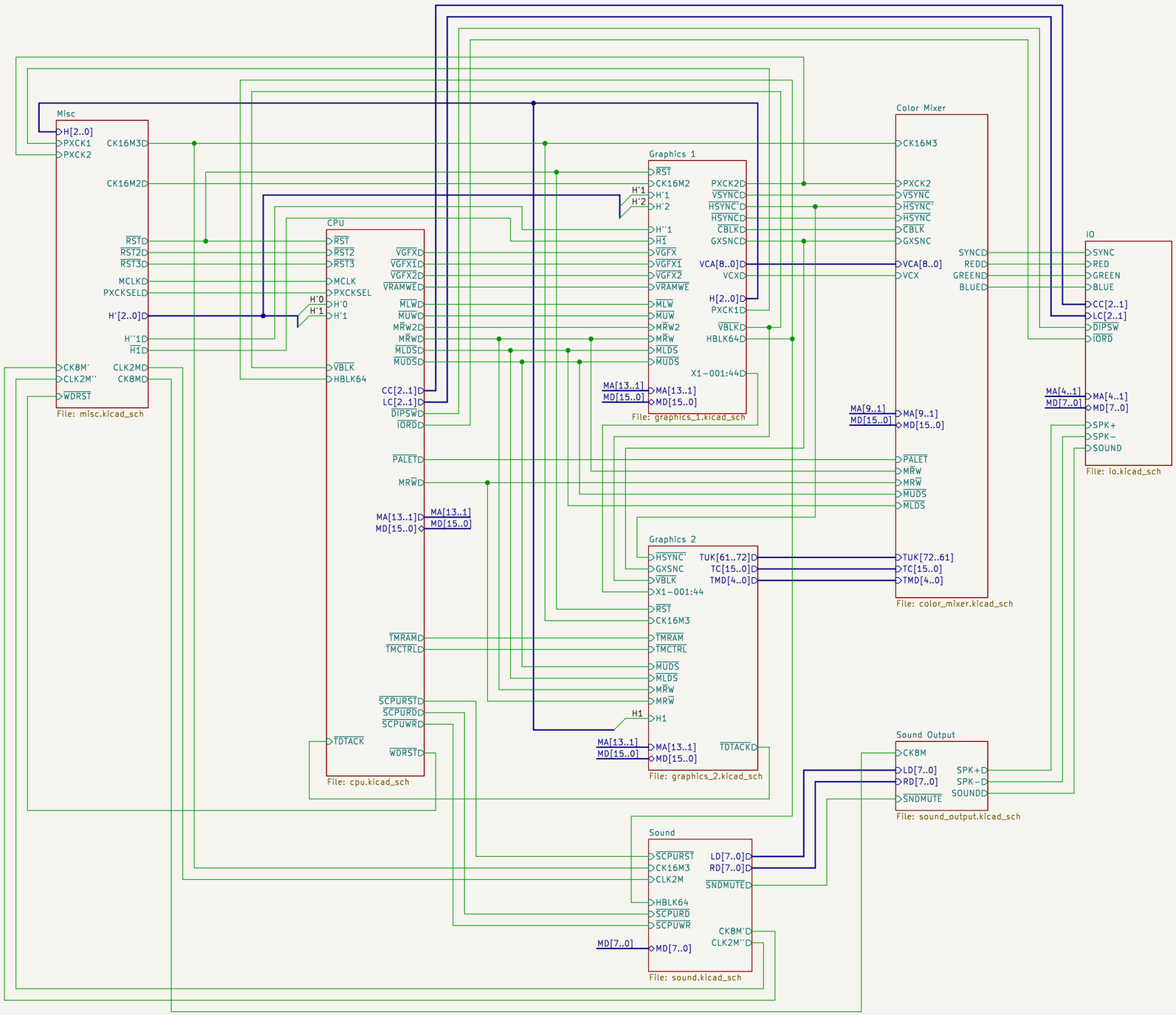


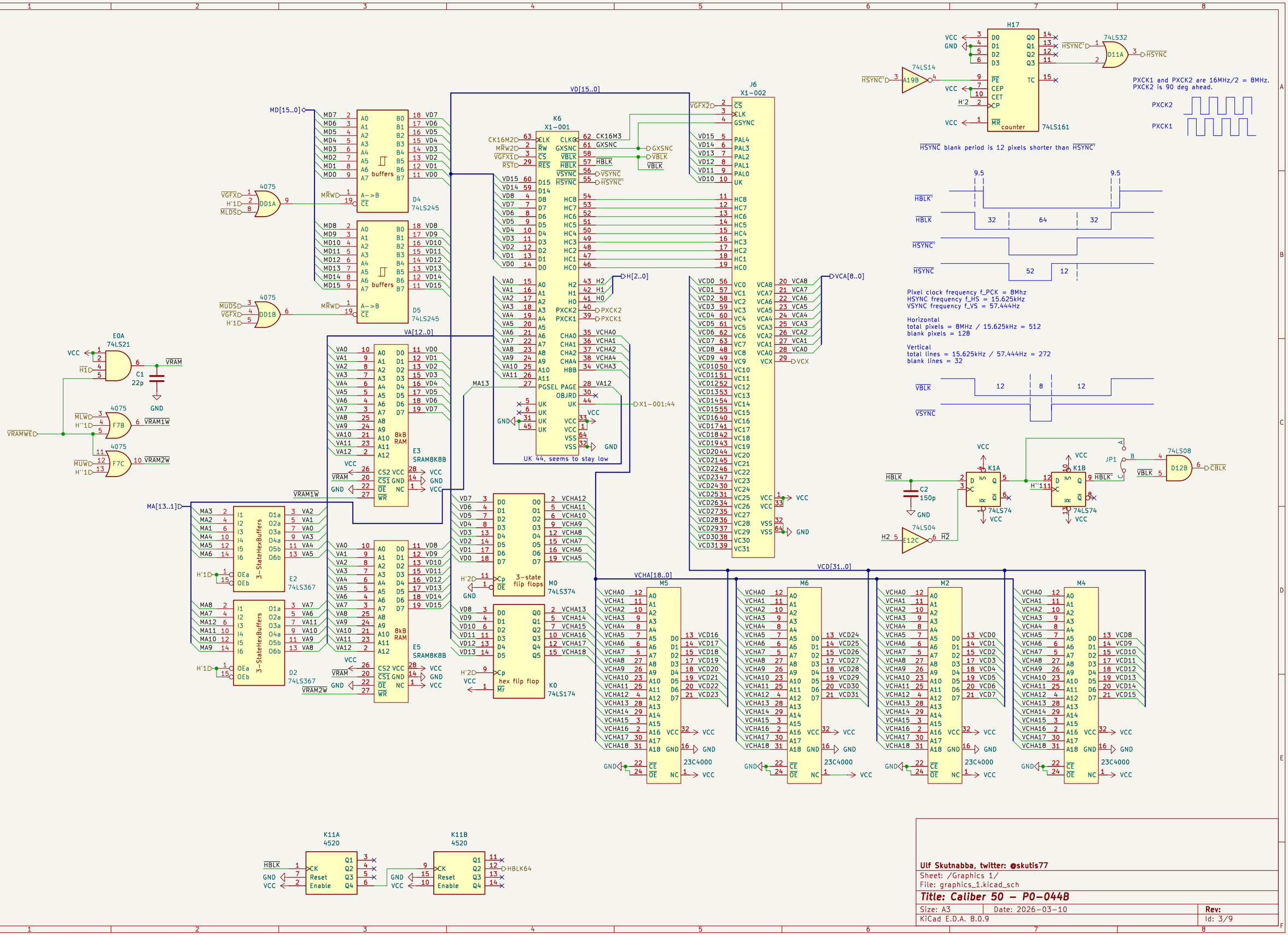
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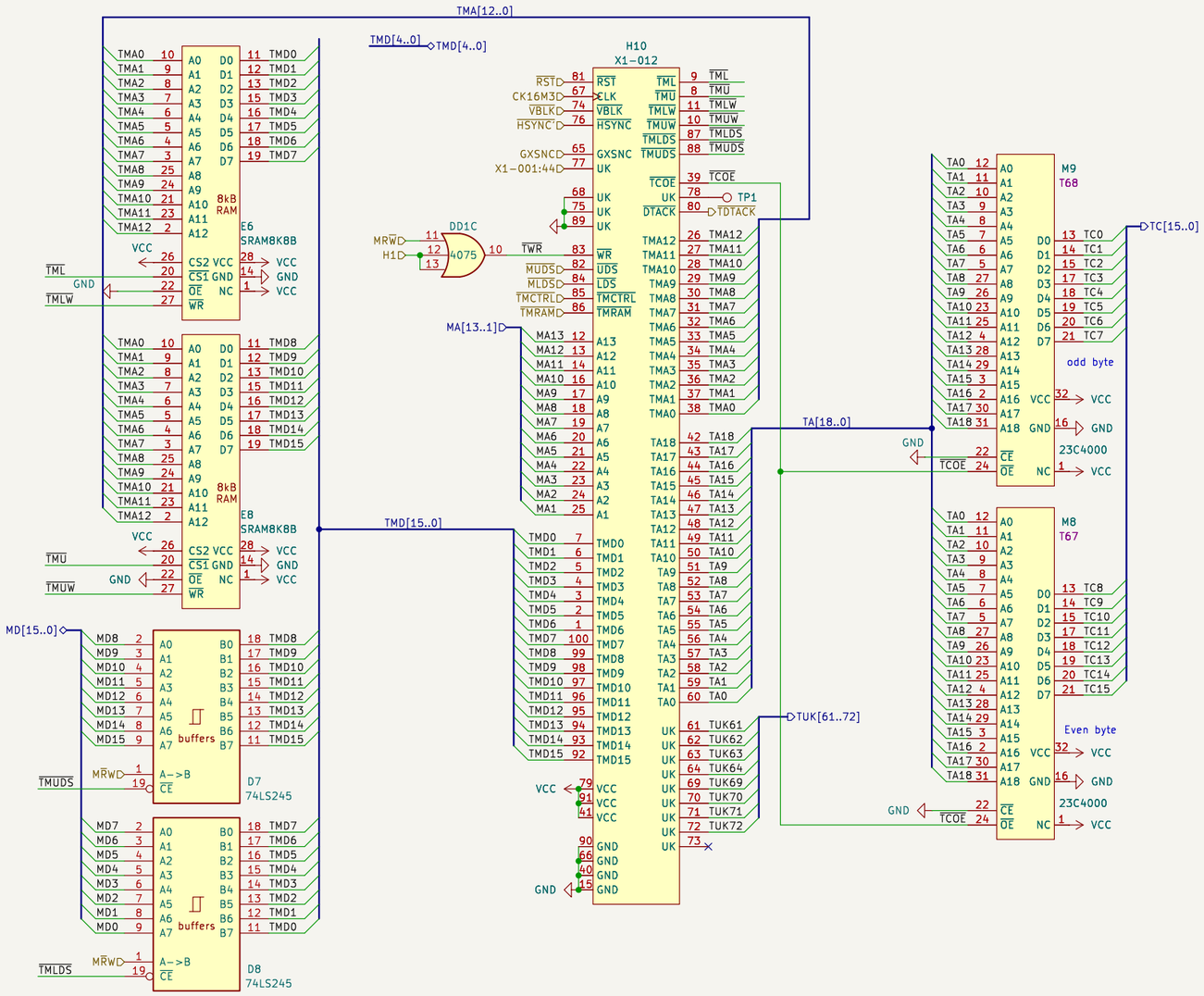
Sheet: /
File: cal50.kicad_sch

Title: Caliber 50 – P0-044B

Size: A3	Date: 2026-03-10	**Rev:**
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B1
MC68000
MD[15..0]
MA[23..1]
MA[13..1]
MCLK
IPL2
IPL1
IPL0
BERR
HALT
RESET
DTACK
BGACK
BR
VPA
VMA
FC0
FC1
FC2
AS
UDS
LDS
RW
BG
MAS
MUDS
MLDS
MRW
RST2
RST
VCC
GND
C0A
C0B
74LS74
HIRST
VIRST
HBLK64
VBLKD
E12F
74LS04
VGFX
F4B
M1C
74LS00
H'1
H'0
PXCKSELL
SNDRW
NVRAM
MRMDTCK
MRAM
PALET
DIPSW
COINCOIL
WDRST
VIRST
NVRAM
HIRST
TDTACK
E10A
74LS133
H2B
MFC
MAS
E11C
74LS32
B6
B5
B11
B12
SRAM32K8B
23C4100
SRAM2K8B
2kB RAM
MA1
MA2
MA3
MA4
MA5
MA6
MA7
MA8
MA9
MA10
MA11
MA12
MA13
MA14
MA15
MA16
MA17
MA18
MA19
MD0
MD1
MD2
MD3
MD4
MD5
MD6
MD7
MD8
MD9
MD10
MD11
MD12
MD13
MD14
MD15
MRAM
MRW'
MLW
MUW
MROML
MROMH
CS
OE
WR
VSS
VCC
GND
NVRAM
RST3
74LS00
CC12C
CC12B
E12D
E12E
74LS04
VCCB
D1
R7
C27
100u
BAT1
Battery_Cell
PWR_FLAG
JP5
JP4
B3
B7
B9
27C512
D10
74LS138
MA19
MA20
MA21
MA22
MA23
MROML
MROMH
decoder
E9
MRMDTCK
HIRST
NVRAM
VIRST
COINCOIL
WDRST
DIPSW
PALET
D9
TMCTRL
TMRAM
IORD
SNDRW
VGFX2
VRAMWE
MRAM
D0B
74LS139
MA9
MA8
VGFX1
E0B
74LS21
D11D
MLDS
F5C
74LS10
VGFX1'
VGFX2
VRAMWE
VGFX
H2A
SCPURD
SCPUWR
K13A
E13C
E11B
E11A
E11D
M1A
D12D
74LS08
74LS32
MRW2
MLW
MUW
MFC is active during interrupts.
F5B
MFC2
MFC1
MFC0
H2D
D6
74LS174
hex flip flop
MD1
MD3
MD5
MD4
MD2
MD0
CC2
LC2
SCPURST
LC1
CC1
CC[2..1]
LC[2..1]
RST
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Sheet: /CPU/
File: cpu.kicad_sch
Title: Caliber 50 – P0-044B
Size: A3
Date: 2026-03-10
KiCad E.D.A. 8.0.9
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VD[15..0]
MD[15..0]
VA[12..0]
VA[18..0]
VCD[31..0]
MA[13..1]
VRAM1W
VRAM2W
VRAMWE
MA13
K6
X1-001
J6
X1-002
H17
74LS161
74LS32
74LS14
74LS04
74LS08
74LS74
H[2..0]
VCA[8..0]
VCX
VGFX2
CK16M2
MRW2
VGFX1
RST
CK16M3
GXSNC
VBLK
HBLK
VSYNC
HSYNC'
HSYNC
PAL4
PAL3
PAL2
PAL1
PAL0
UK
HC8
HC7
HC6
HC5
HC4
HC3
HC2
HC1
HC0
PXCK2
PXCK1
VCHA0
VCHA1
VCHA2
VCHA3
VCHA4
VCHA5
VCHA6
VCHA7
VCHA8
VCHA9
VCHA10
VCHA11
VCHA12
PGSEL
PAGE
OBJRD
HBB
VC0
VC1
VC2
VC3
VC4
VC5
VC6
VC7
VC8
VC9
VC10
VC11
VC12
VC13
VC14
VC15
VC16
VC17
VC18
VC19
VC20
VC21
VC22
VC23
VC24
VC25
VC26
VC27
VC28
VC29
VC30
VC31
UK 44, seems to stay low
PXCK1 and PXCK2 are 16MHz/2 = 8MHz.
PXCK2 is 90 deg ahead.
HSYNC blank period is 12 pixels shorter than HSYNC'
9.5
32
64
52
12
Pixel clock frequency f_PCK = 8Mhz
HSYNC frequency f_HS = 15.625kHz
VSYNC frequency f_VS = 57.444Hz
Horizontal
total pixels = 8MHz / 15.625kHz = 512
blank pixels = 128
Vertical
total lines = 15.625kHz / 57.444Hz = 272
blank lines = 32
VBLK
VSYNC
12
8
12
C1
22p
C2
150p
JP1
CBLK
D1A
D2A
D3A
D4A
D5A
D6A
D7A
D8A
D9A
D10A
D11A
D12A
D1B
D2B
D3B
D4B
D5B
D6B
D7B
D8B
D9B
D10B
D11B
D12B
D1C
D2C
D3C
D4C
D5C
D6C
D7C
D8C
D9C
D10C
D11C
D12C
E0A
E1A
E2A
E3A
E4A
E5A
E6A
E7A
E8A
E9A
E10A
E11A
E12A
E13A
E14A
E15A
E16A
E17A
E18A
E19A
E20A
E21A
E22A
E23A
E24A
E25A
E26A
E27A
E28A
E29A
E30A
E31A
E32A
E33A
E34A
E35A
E36A
E37A
E38A
E39A

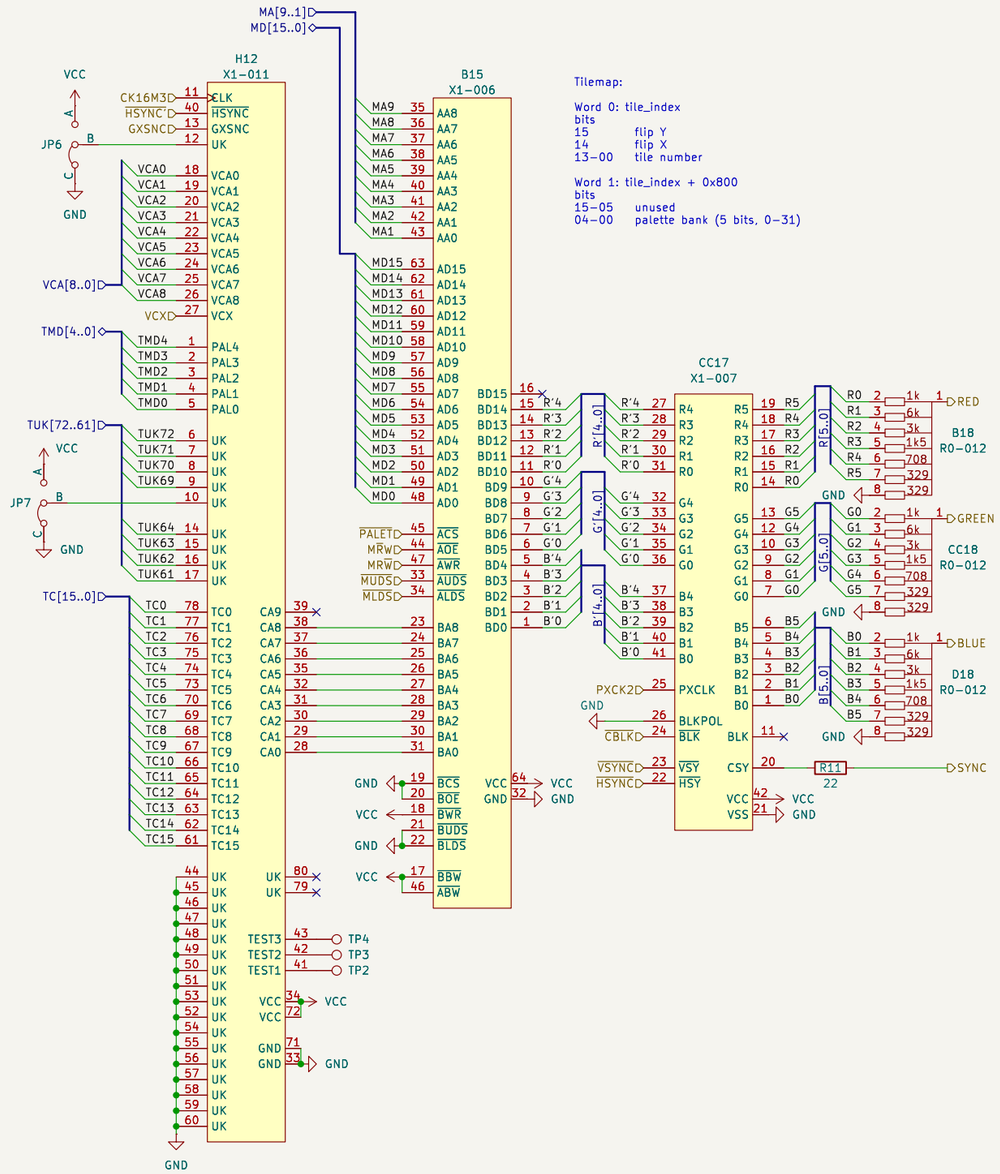


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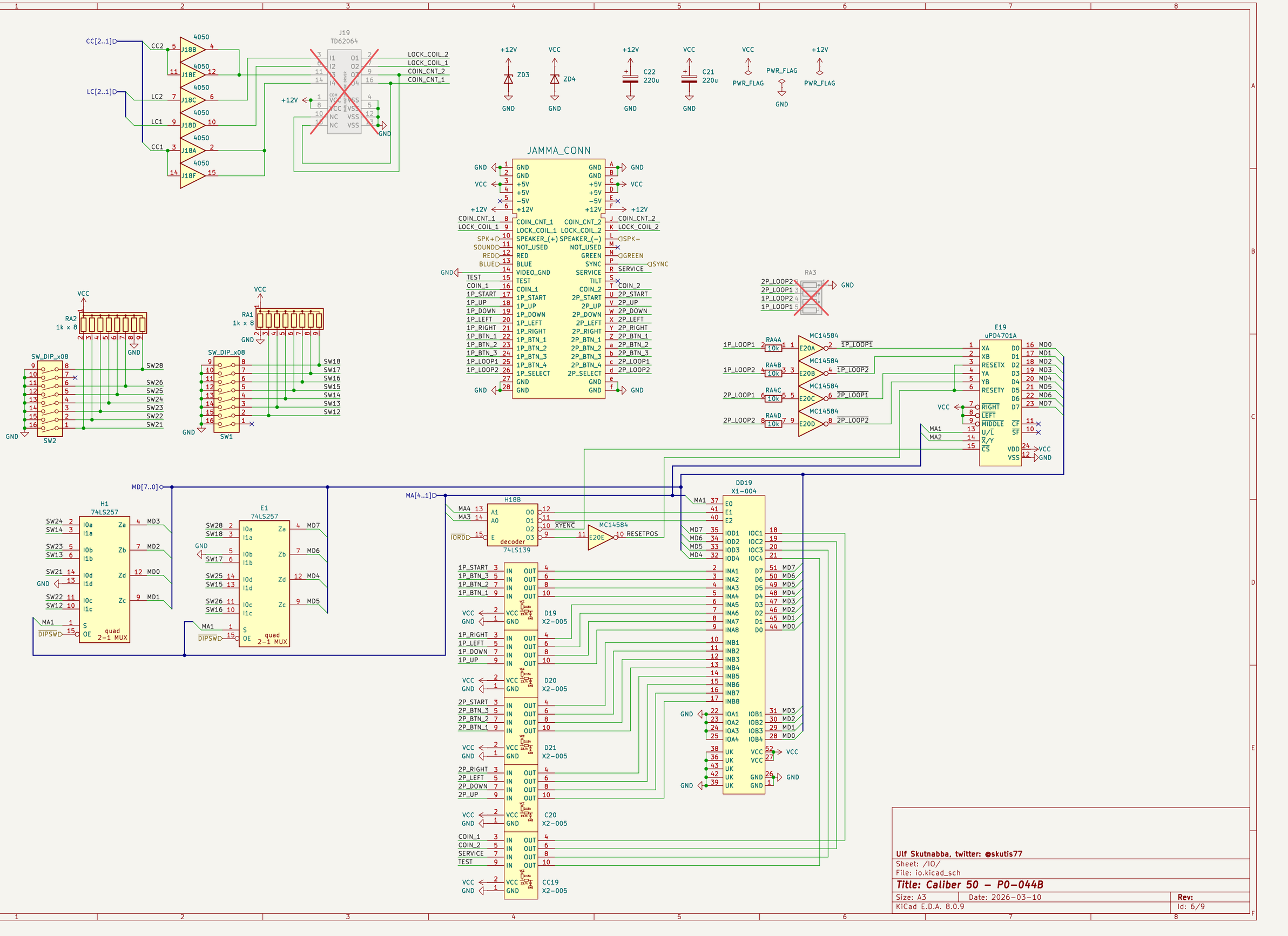
Sheet: /Graphics 2/
File: graphics_2.kicad_sch

Title: Caliber 50 – P0-044B

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Sheet: /Color Mixer/	
File: color_mixer.kicad_sch	
Title: Caliber 50 – P0-044B	
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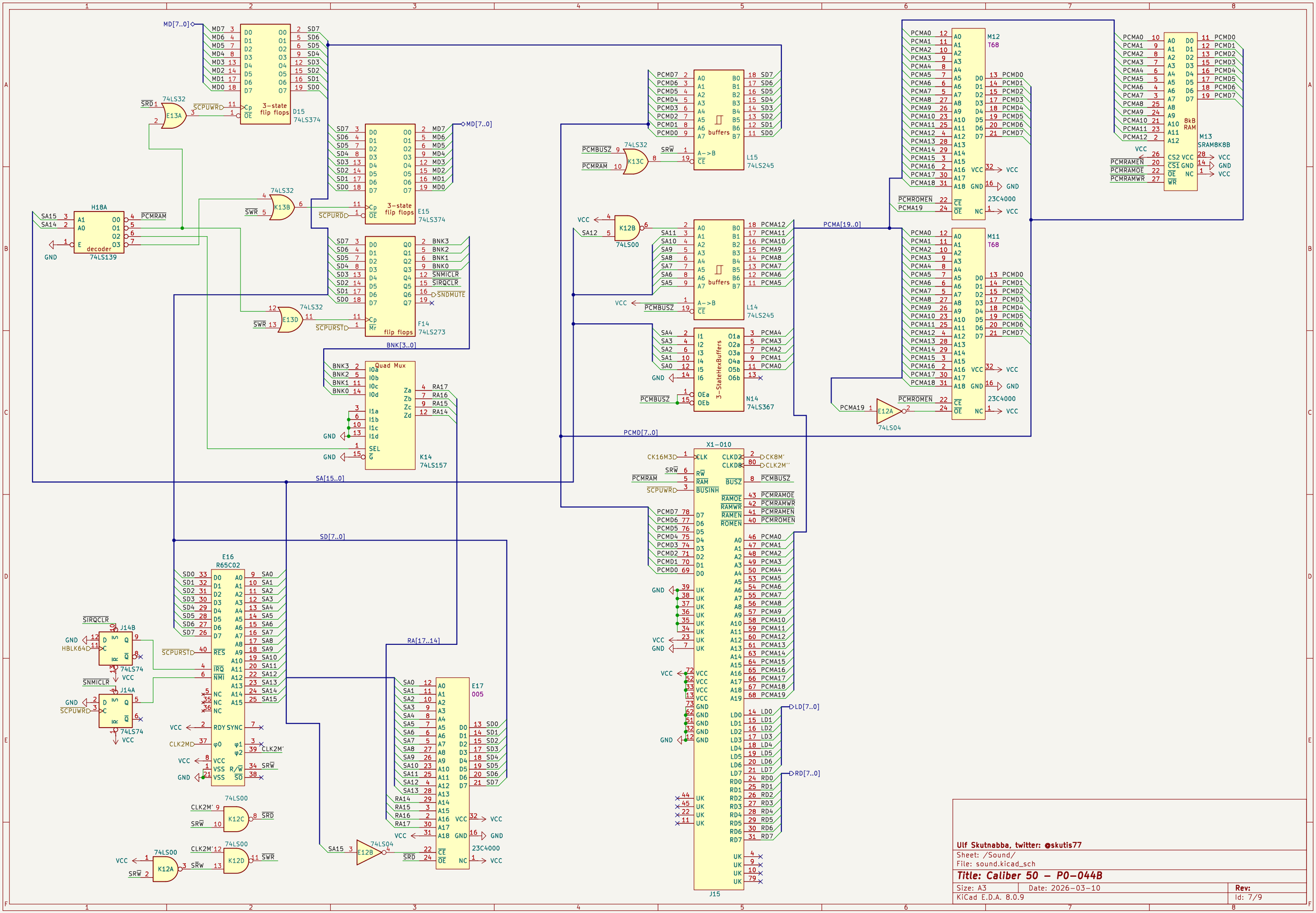
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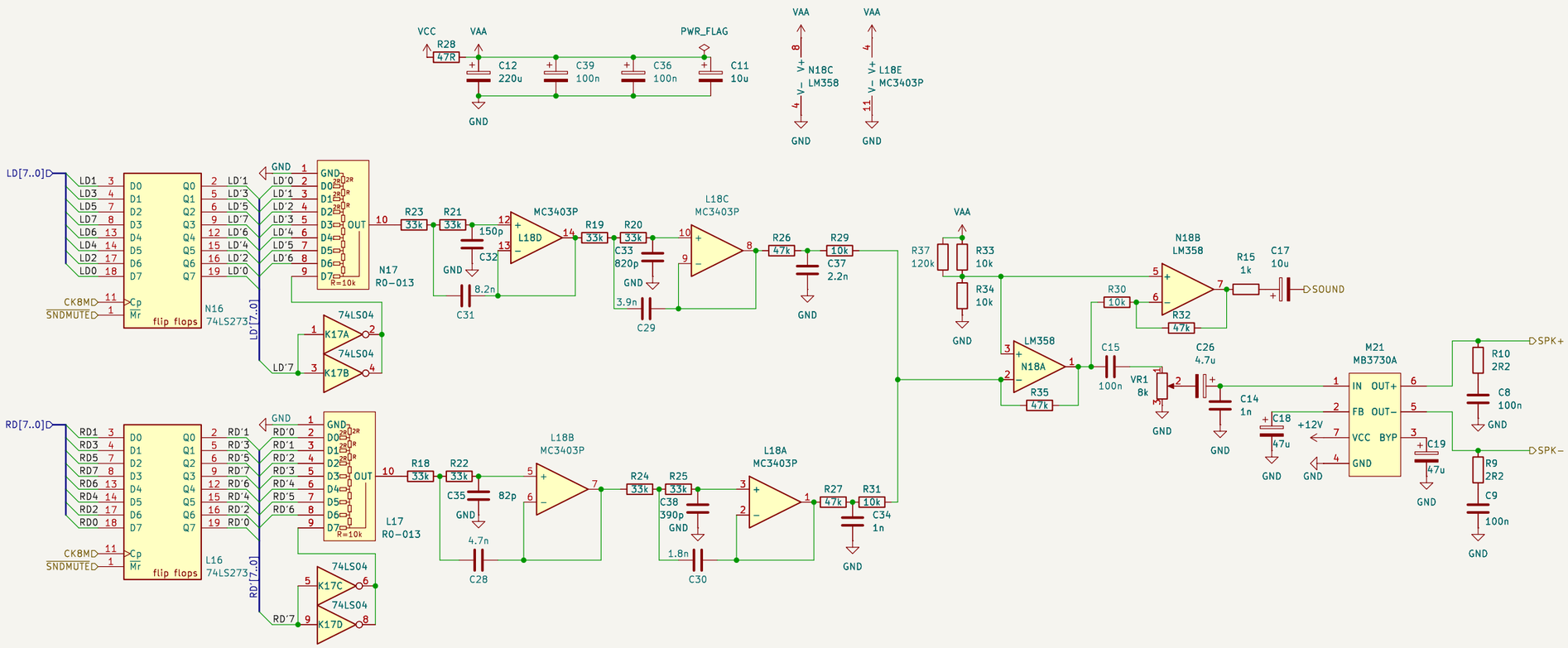
Sheet: /IO/

File: io.kicad_sch

Title: Caliber 50 – P0-044B

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MD[7..0]
SD[7..0]
SA[15..0]
BNK[3..0]
RA[17..14]
PCMA[19..0]
PCMD[7..0]
LD[7..0]
RD[7..0]
D15 74LS374
E15 74LS374
F14 74LS273
E13A 74LS32
K13B 74LS32
E13D 74LS32
H18A decoder 74LS139
K14 Quad Mux 74LS157
L15 buffers 74LS245
K13C 74LS32
K12B 74LS00
L14 buffers 74LS245
N14 3-StateHexBuffers 74LS367
M12 T68
M11 T68
M13 SRAM8K8B 8kB RAM
23C4000
E12A 74LS04
X1-010
J15
E16 R65C02
J14B 74LS74
J14A 74LS74
K12C 74LS00
K12D 74LS00
K12A 74LS00
E17 005
E12B 74LS04
SIRQCLR
SNMICLR
SCPURST
SCPURD
SCPUWR
SRD
SWR
SRW
PCMBUSZ
PCMRAM
PCMROMEN
PCMRAMEN
PCMRAMOE
PCMRAMWR
PCMA19
CK16M3
CK8M'
CLK2M''
CLK2M'
HBLK64
SNDMUTE
VCC
GND
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Sheet: /Sound/
File: sound.kicad_sch
Title: Caliber 50 – P0-044B
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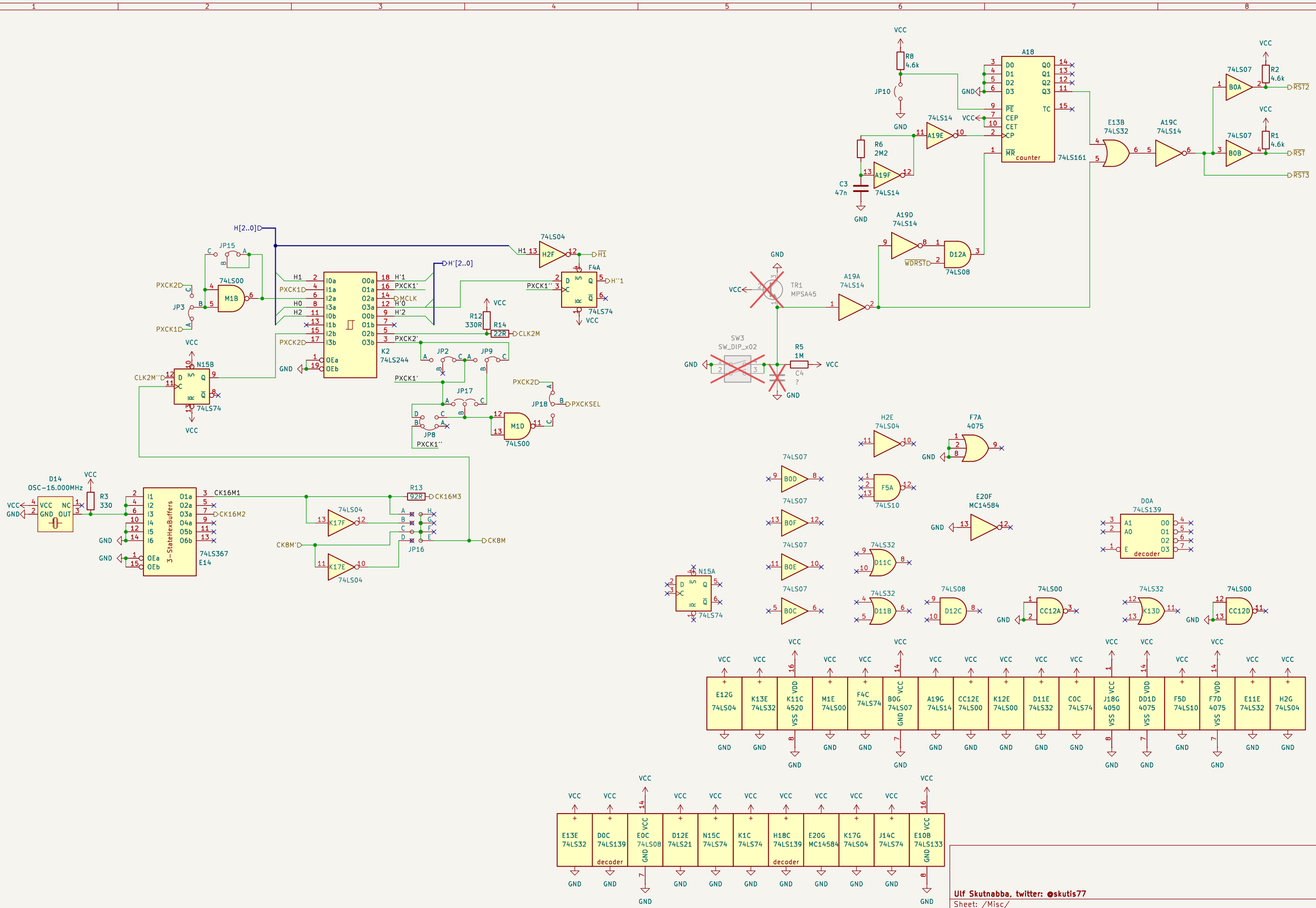


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Sheet: /Sound Output/
File: sound_output.kicad_sch

Title: Caliber 50 – P0-044B

Size: A3	Date: 2026-03-10	**Rev:**
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H[2..0]
JP15
PXCK2
PXCK1
JP3
74LS00
M1B
H1
H0
H2
I0a
I1a
I2a
I3a
I0b
I1b
I2b
I3b
OEa
OEb
K2
74LS244
O0a
O1a
O2a
O3a
O0b
O1b
O2b
O3b
H'1
PXCK1'
MCLK
H'0
H'2
PXCK2'
H'[0..2]
74LS04
H2F
F4A
PXCK1''
H''1
74LS74
R12
330R
R14
22R
CLK2M
JP2
JP9
JP17
JP18
PXCKSEL
JP8
M1D
74LS00
N15B
CLK2M''
74LS74
D14
OSC-16.000MHz
R3
330
CK16M1
CK16M2
74LS367
E14
74LS04
K17F
K17E
R13
92R
CK16M3
JP16
CK8M'
CK8M
VCC
GND
R8
4.6k
JP10
A18
D0
D1
D2
D3
Q0
Q1
Q2
Q3
PE
CEP
CET
CP
MR
TC
counter
74LS161
74LS14
A19E
A19F
R6
2M2
C3
47n
E13B
74LS32
A19C
74LS14
74LS07
B0A
B0B
R2
4.6k
R1
4.6k
RST12
RST
RST13
A19D
74LS14
D12A
74LS08
WDRST
TR1
MPSA45
A19A
74LS14
SW3
SW_DIP_x02
R5
1M
C4
?
H2E
74LS04
F7A
4075
74LS07
B0D
B0F
B0E
B0C
F5A
74LS10
74LS32
D11C
D11B
E20F
MC14584
D0A
74LS139
A1
A0
E
O0
O1
O2
O3
decoder
74LS08
D12C
74LS00
CC12A
CC12D
74LS32
K13D
N15A
74LS74
E12G 74LS04
K13E 74LS332
K11C 4520
M1E 74LS00
F4C 74LS74
B0G 74LS07
A19G 74LS14
CC12E 74LS00
K12E 74LS00
D11E 74LS32
C0C 74LS74
J18G 4050
DD1D 4075
F5D 74LS10
F7D 4075
E11E 74LS32
H2G 74LS04
E13E 74LS32
D0C 74LS139
E0C 74LS08
D12E 74LS21
N15C 74LS74
K1C 74LS74
H18C 74LS139
E20G MC14584
K17G 74LS04
J14C 74LS74
E10B 74LS133
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Sheet: /Misc/
File: misc.kicad_sch
Title: Caliber 50 – P0-044B
Size: A3
Date: 2026-03-10
Rev:
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